Program 1: Understanding Constructor Chaining and Method Overloading

📌 Instructions:

Analyse the code (in Program 1) below carefully. Then answer the questions that follow.

```
class Learn {
  Learn() {
    this("Smart");
    System.out.println("Default Constructor");
  }

  Learn(String x) {
    System.out.println("Learn, " + x);
  }

  void show(String x) {
    System.out.println("Show: " + x);
  }

  void show(String x, String y) {
    System.out.println("Show: " + x + ", " + y);
  }

  public static void main(String[] args) {
    Learn obj = new Learn();
    obj.show("Focus");
  }
}
```

Program 1 : Learn.java

❓ Questions:

1. Which constructor is called first when `new Learn()` is created?
2. What is **constructor chaining** in this code? Which line shows this?
3. Identify which part of the code shows **method overloading**.
4. How many times is the **constructor invoked** when you run the program?
5. What is the **output** of this program?

1. The Learn(String x) constructor
2. Constructor chaining is a constructor calling another constructor in the same class using 'this'
3. void show (String x) and void show (String x, String y)
4. 2 times
5. Learn, Smart
   Default Constructor
   Show: Focus

Program 2 : Analysing Constructor and Method Overloading

Instructions:

Read the code below carefully. **Do not run the code.** Analyse the structure and answer the questions that follow.

```
class Calculator {
  int a, b;

  Calculator(int x, int y) {
    a = x;
    b = y;
    System.out.println("[Constructor] Sum: " + (a + b));
  }

  Calculator() {
    this(20, 10);
    System.out.println("[Constructor] Difference: " + (a - b));
  }

  int calculate(int x, int y) {
    return x + y;
  }

  int calculate(int x, int y, int z) {
    return x + y + z;
  }

  public static void main(String[] args) {
    Calculator calc = new Calculator();

    int result1 = calc.calculate(5, 15);
    int result2 = calc.calculate(1, 2, 3);

    System.out.println("[Method] Sum of 5 and 15: " + result1);
    System.out.println("[Method] Sum of 1, 2, and 3: " + result2);
  }
}
```

Program 2 : Calculator.java

? Questions:

1. How many constructors are defined in the class?
2. Which constructor is called first when `new Calculator()` is used?
3. What is the purpose of `this(20, 10);` in the default constructor?
4. How is **constructor chaining** applied in this program?
5. Identify the **method overloading** in this code.
6. How many times are methods named `calculate` executed in `main()`?
7. What is the expected output of the program?

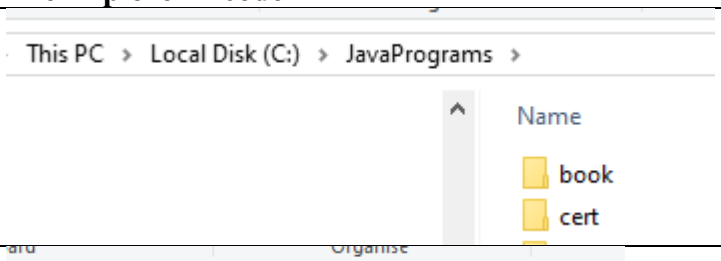
7. [Constructor] Sum: 30
   [Constructor] Difference: 10
   [Method] Sum of 5 and 15: 20
   [Method] Sum of 1, 2 and 3: 6

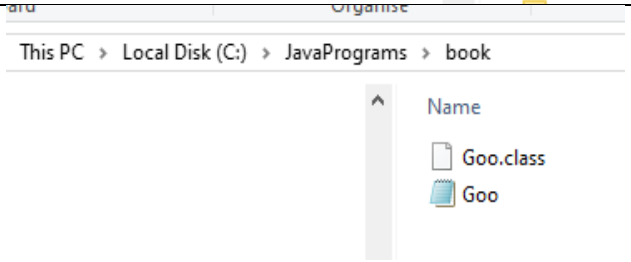
Concepts Covered:

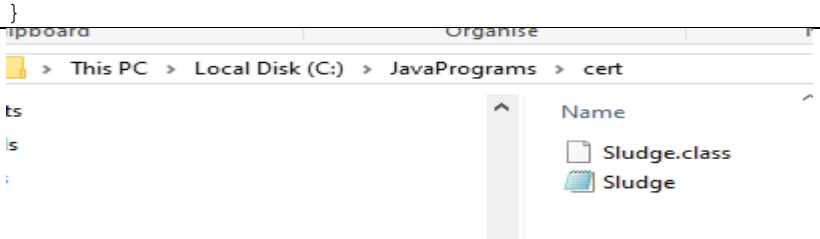
1. Two constructors
2. Calculator(int x, int y) because it called by this(20,30)
3. Perform constructor chaining by calling another constructor in the same class
4. Using this(20,30) to call another constructor
5. int calculate(int x,int y) and int calculate(int x,int y,int z)
6. Two times

- Constructor chaining with `this(...)`
- Method overloading based on parameters
- Arithmetic operations (add/subtract)
- Execution flow of constructors

Program 3 : Package (refer Lec03AnatomyJava slide 27-28)

#	**File Explorer + code**
1	

2	
	`package book;`
	
	`import cert.Sludge;`
	`// Import the Sludge class from the cert package`
	
	`public class Goo {`
	`  public static void main(String[] args) {`
	`  Sludge o = new Sludge();`
	`  o.testIt();`
	`  }`
	`}`

3	
	`package cert;`
	
	`public class Sludge {`
	`  public void testIt() {`
	`  System.out.println("sludge");`
	`  }`
	`}`

Program 4 : Carpet room case study

Below is a complete **Java mini project** that aligns perfectly with **Chapter 2 topics**: classes, objects, constructors, methods (including `toString`, `equals`), and passing parameters by object.

Exercise Title: *Calculating Carpet Cost for Hotel Rooms*

Objective:

By completing this exercise, students will be able to:

- Create and use **classes and objects**.
- Implement and use **constructors**.
- Write and call **methods**, including `toString()` and `equals()`.
- Pass **objects as parameters** to other classes.
- Apply **object-oriented programming** concepts in a real-world scenario.

Instructions:

◆ Step 1: Create a `Room` class

This class will store the room's length and width, calculate its area, and compare rooms.

```
public class Room {
    private double length;
    private double width;

    public Room(double length, double width) {
        this.length = length;
        this.width = width;
    }

    public double getArea() {
        return length * width;
    }

    @Override
    public String toString() {
        return "Room [Length = " + length + ", Width = " + width + ", Area
= " + getArea() + " sq. meters]";
    }

    public boolean equals(Room otherRoom) {
        return this.getArea() == otherRoom.getArea();
    }
}
```

◈ Step 2: Create a `Carpet` class

This class stores the carpet price per square meter.

```
public class Carpet {
    private double pricePerSqMeter;

    public Carpet(double price) {
        this.pricePerSqMeter = price;
    }

    public double getPrice() {
        return pricePerSqMeter;
    }

    @Override
    public String toString() {
        return "Carpet [Price per square meter = RM " + pricePerSqMeter +
"]";
    }
}
```

◈ Step 3: Create a `RoomCarpet` class

This class combines a room and a carpet object, calculates total carpet cost.

```
public class RoomCarpet {
    private Room room;
    private Carpet carpet;

    public RoomCarpet(Room room, Carpet carpet) {
        this.room = room;
        this.carpet = carpet;
    }

    public double getTotalCost() {
        return room.getArea() * carpet.getPrice();
    }

    @Override
    public String toString() {
        return room.toString() + "\n" + carpet.toString() + "\nTotal Carpet
Cost = RM " + getTotalCost();
    }
}
```

◈ Step 4: Create a `Main` class to run the program

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        // Get room dimensions
```

```
        // Get carpet price

        // Create objects



        // Display details
        System.out.println("\n=== Room Carpet Summary ===");



        // Testing equals method



        System.out.println("\nIs this room equal to another room with the
same size?");



// should return true

        input.close();
    }
}
```

✅ Expected Output (Sample)

```
Enter room length (in meters): 5
Enter room width (in meters): 4
Enter carpet price per square meter: RM 12.5

=== Room Carpet Summary ===
Room [Length = 5.0, Width = 4.0, Area = 20.0 sq. meters]
Carpet [Price per square meter = RM 12.5]
Total Carpet Cost = RM 250.0

Is this room equal to another room with the same size?
true
```

📚 Topics Covered from Chapter 2:

- ✅ Object & Class creation
- ✅ Constructor & parameterized constructors
- ✅ Method overloading
- ✅ `toString()` method
- ✅ `equals()` method
- ✅ Passing objects as parameters
- ✅ Basic encapsulation with private variables
